

Partial Differential Equations

Defⁿ:- A Equation which involves a dependent variable and its derivations with respect to two or more independent variables is called partial differential eqⁿ.

eg $x \cdot \frac{\partial z}{\partial y} + 4y \frac{\partial z}{\partial x} = 2z + 3xy$

Notations:- $\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial^2 z}{\partial x^2} = r,$

$\frac{\partial^2 z}{\partial x \partial y} = s, \frac{\partial^2 z}{\partial y^2} = t$

Linear & non linear Partial differential

Eqⁿ (P.D.E):-

if the partial ~~deriv~~ derivatives of the dependent variable occur in first degree only & not multiplied together such a P.D.E is said to be linear P.D.E, otherwise it is said to be non-linear P.D.E.

Formation of Partial d.e.s:-

Partial d.e.s can be formed by two methods. :-

1) By the elimination of arbitrary constants.

2) By the elimination of arbitrary functions.

① By elimination of arbitrary constants:-

Let the given fⁿ be $f(x, y, z, a, b) = 0$ where a & b are arbitrary constants. ①

To eliminate a & b , differentiate ① Partially w.r.t x & y .

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot p = 0 \quad \text{--- ②}$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot q = 0 \quad \text{--- ③}$$

Now eliminate the constants a & b

from ①, ② & ③, we get a

Partial d.e.s of the first order of the form $\phi(x, y, z, p, q) = 0$.

① Form the Partial eqⁿ by eliminating the arbitrary constants a & b from $Z = ax + by + ab$

Sol

$$Z = ax + by + ab \text{ ——— ①}$$

diff ① partially w.r.t x & y , we get-

$$\frac{\partial Z}{\partial x} = a \Rightarrow b = a \text{ ——— ②}$$

$$\frac{\partial Z}{\partial y} = b \Rightarrow a = b \text{ ——— ③}$$

Putting the value of a & b in ①

$$\boxed{Z = bx + ay + b^2}$$

$$27 \quad Z = ax + by + a^2 + b^2$$

Sol

$$\frac{\partial Z}{\partial x} = a \Rightarrow b = a$$

$$\frac{\partial Z}{\partial y} = b \Rightarrow a = b$$

$$\Rightarrow Z = bx + ay + b^2 + a^2$$

3) $Z = (x+a)(y+b)$

form the P.d.eqn by eliminating the arbitrary constants.

Sol

$$Z = (x+a)(y+b)$$

$$Z = xy + bx + ay + ab$$

$$\frac{\partial Z}{\partial x} = y + b \quad \text{--- (1)}$$

$$\frac{\partial Z}{\partial y} = a + x \quad \text{--- (2)}$$

putting these values in the given eqn

$$\boxed{Z = px}$$

c) Solve

$$x^2 + y^2 + (z-c)^2 = a^2,$$

$$x^2 + y^2 + (z-c)^2 = a^2 \quad \text{--- (1)}$$

Sol

Given

Partially w.r.t x & y .

$$\text{diff (1) Partially w.r.t } x \Rightarrow 2x + 0 + 2(z-c) \frac{\partial z}{\partial x} = 0$$

$$x + (z-c) \frac{\partial z}{\partial x} = 0$$

$$x + (z-c)p = 0$$

$$(z-c)p = -x$$

$$\rightarrow (z-c) = -\frac{x}{b} \quad \text{--- (2)}$$

Now diff (1) w.r.t y

$$2y + 2(z-c) \cdot \frac{\partial z}{\partial y} = 0$$

$$y + (z-c)z = 0$$

$$(z-c)z = -y$$

$$(z-c) = -\frac{y}{z} \quad \text{--- (3)}$$

$\rightarrow$ Now from (2) Δ (3)

$$y + \left(-\frac{x}{b}\right) \cdot z = 0$$

$$yb - xz = 0 \quad \text{which is the}$$

required p. deg n.

5) Q Remove the p. deg n by eliminating arbitrary constants.

$$(x-a)^2 + (y-b)^2 + z^2 = r^2$$

67 Form the Pdeqn

$$\log(az-1) = x+ay+b.$$

Sol

$$\log(az-1) = x+ay+b \text{ ——— (1)}$$

Formation of Partial deqn by the elimination of arbitrary functions.

Sol For the formation of Pdeqn, we have to eliminate the fns.

Q ① Form a Pdeqn by eliminating the arbitrary fns

$$z = f(x^2 + y^2)$$

Sol $z = f(x^2 + y^2)$ — (1)

diff ① Partially w.r.t x

$$\frac{\partial z}{\partial x} = f'(x^2 + y^2) \cdot 2x \text{ — (2)}$$

diff ① Partially w.r.t y .

$$\frac{\partial z}{\partial y} = f'(x^2 + y^2) \cdot 2y \text{ — (3)}$$

Divide (2) & (3)

$$\frac{p}{q} = \frac{2x}{2y}$$

$$\boxed{y p - x q = 0}$$

Required Partial deqn.

diff ① Partially w.r.t x & y

$$\begin{aligned} \rightarrow yz + xy \cdot \frac{\partial z}{\partial x} &= f'(x^2 + y^2 + z^2) \cdot 2x + 2z \cdot \frac{\partial z}{\partial x} \\ \Rightarrow yz + xy p &= f'(x^2 + y^2 + z^2) \cdot 2x. \end{aligned}$$

Q ② $xyz = f(x^2 + y^2 + z^2)$

Sol Given $xyz = f(x^2 + y^2 + z^2)$ — ①

diff ① Partially w.r.t x & y .

$$x \cdot \left[y \frac{\partial z}{\partial x} \right] + yz \cdot 1 = f'(x^2 + y^2 + z^2) \cdot \left(2x + 2z \frac{\partial z}{\partial x} \right)$$

$$yz + yx p = f'(x^2 + y^2 + z^2) (2x + 2z p) \quad \text{--- ②}$$

$$xz + xy \cdot q = f'(x^2 + y^2 + z^2) (2y + 2z q) \quad \text{--- ③}$$

divide ② by ③

$$\frac{yz + xyp}{xz + xyq} = \frac{2x + 2zp}{2y + 2zq}$$

$$(yz + xyp)(2y + 2zq) = (xz + xyq)(2x + 2zp)$$

$$\Rightarrow (yz + xyp)(y + zq) = (xz + xyq)(x + zp)$$

$$\Rightarrow y^2z + yz^2q + xy^2p + xyzpq = x^2z + xz^2p + x^2yq + xyzpq$$

$$\Rightarrow y^2z + yz^2q + xy^2p = x^2z + xz^2p + x^2yq$$

$$\Rightarrow x(y^2 - z^2)p + y(z^2 - x^2)q = (x^2 - y^2)z$$

required Pdeqn.

$$3) \quad xyz = f(x + y + z)$$

$$4) \quad x = f(x+at) + g(x-at)$$

form the pdeqn by eliminating the arbitrary fns.